Candidate seat no:______________

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

Fifth Semester of B.Tech. Theory (IT) Examination

November 2013

IT304.01 Modeling Simulation & Operational Research

Date: 27/11/2013,Wednesday Time: 10:00 a.m. to 01:00 p.m. Maximum Marks:70

Instructions:

1. **This question paper comprises of two sections.**
2. **Section I and II must be attempted in separate answer sheets.**
3. **Make suitable assumptions and draw neat figures wherever if required.**
4. **Use of scientific non-programable calculator is allowed.**

SECTION I

Q.1. Answer the following questions. [07]

(a) What do you understand by Optimization Problem? [04]

(b) Discuss: how Operations Research can be helpful to an engineer? [03]

Q.2. Answer the following questions. [14]

(a) Write whether following statements are true or false [04]

i. In graphical solution of linear programming problem, any point which is a feasible solution lie in the first quadrant.

ii. One of the properties of Linear Programming Problem is that The relationship between problem variables and constraints must be linear

iii. In graphical solution of solving Linear Programming problem to convert inequalities into equations, we use Slack variables

iv. To convert $\geq$ type of inequality in LPP into equations, we have to subtract slack Variable

(b) Solve the following LPP using graphical method. [06]
Maximize $Z = 5x_1 + 3x_2$, Subject to
$3x_1 + 5x_2 \leq 15$, $5x_1 + 2x_2 \leq 10$, $x_1 \geq 0, x_2 \geq 0$

(c) Define the following terms in LPP. [04]

i. solution

ii. feasible solution

iii. optimal feasible solution

iv. infeasible solution

OR

Q.2. Answer the following questions. [14]

(a) Write the dual LPP of following linear programming problem. [04]
Maximize $Z = 4x_1 + 10x_2$, Subject to
$2x_1 + x_2 \leq 6$, $3x_1 + 5x_2 \leq 19$, $2x_1 + 3x_2 \geq 11$, $x_1 \geq 0, x_2 \geq 0$

(b) GNFC chemicals manufactures two products say, naptha (x) and urea (y) in its plant located near by Baroda. It gets a profit of Rs. 70 per unit for naptha and Rs. 100 per unit for urea. The time requirements for each product and the total available time in each department are shown in the following table. Each product has to pass through two departments in the plant. [04]

Department	Time requires per unit (hours)		Hours available in a month
	Neptha	Urea	
1	3	4	2100
2	4	3	2100

Further, the demand for these products restricts the production to a maximum of 450 units of each product. Formulate this problem into a linear programming problem (Do not solve it)

(c) Solve the following LPP by using Big-M method. [06]
Maximize $Z = 4.5x_1 + 15x_2$, Subject to
$3x_1 + 1.5x_2 \leq 48$, $4.5x_1 + 6x_2 \leq 120$, $x_1 \geq 12$, $x_2 \geq 15$, $x_1 \geq 0, x_2 \geq 0$

Q.3. Answer the following questions. [14]

(a) Write whether following statements are true or false. [04]

i. When the total allocation in a transportation problem with m rows and n columns is not equal to $m+n-1$ the situation is known as degeneracy.

ii. In Northwest corner method the allocation are made starting from the left hand side top corner.

iii. In an Assignment problem, the horizontal and vertical lines drawn to cover all zeros of total opportunity matrix must be equal to $m \times n$ (where m and n are number of rows and columns respectively)

iv. For a balanced transportation problem total supply is equal to total demand.

(b) Define Integer Programming Problem. Write name of methods of solving integer programming problem. [03]

(c) Machineco has four machines and four jobs to be completed.Each machine must be assigned to complete one job. The time required to set up each machine for completing each job is shown in Table below.Deternine assignment of machines to minimize the total setup time needed to complete the four jobs. Also write the optimum setup time. [07]

Machine	Time(hours) Job1	Job2	Job3	Job4
1	14	5	8	7
2	2	12	6	5
3	7	8	3	9
4	2	4	6	10

OR

Q.3. Answer the following questions. [14]

(a) Define Assignment prblem. Write the formulation as Linear Programming Model. [04]

(b) What is unbalanced transportation problem? Write the procedure to obtain balanced transportation problem. [04]

(c) Find initial basic feasible solution to following transportation problem using north west corner method. (Entries in table represent unit cost of transportation)Also determine the total cost of transportation. [06]

	D_1	D_2	D_3	D_4	Supply
S_1	3	7	6	4	5
S_2	2	4	3	2	2
S_3	4	3	8	5	3
Demand	3	3	2	2	

SECTION II

Q.4. Answer the following question. [07]

(a) Write the correct option from following. [04]

i. In Queuing theory traffic intensity is given by

A. $\frac{\lambda}{\mu}$ B. $\frac{\mu}{\lambda}$ C. $\lambda \times \mu$ D. None of these

Where λ denotes arrival rate and μ denotes service rate respectively.

ii. Dijkstra's algorithm is designed to determine the shortest routes between source node and every other node in the network. A. True B. False

iii. In PERT, the optimistic time estimate of duration of an activity is considered when execution goes extremely poor. A. True B. False

iv. In Kendall's notation $\{(a/b/c) : (d/e)\}$of queuing system the symbol c represent number of servers. A. True B. False

(b) What is simulation? Explain in brief. [03]

Q.5. Answer the following questions. [14]

(a) Explain in brief the following terms in queuing system. [07]

i. Arrival pattern
ii. Service mechanism or service pattern
iii. Queue discipline
iv. Customer behavior

(b) The arrival at a telephone booth is considered to be following Poisson distribution with an average of 10 minutes between one arrival and the next. Length of the phone call is assumed to be distributed exponentially with a mean of 5 minutes.Answer the following questions, [07]

i. What is the probability that a person arriving at the booth will have to wait?
ii. What is the average length of queue that forms time to time?
iii. What is average waiting time of person in system?
iv. What is the probability that there are more than 2 customers in system?

OR

Q.5. Answer the following questions. [14]

(a) Define the following terms with reference to queuing system. [07]

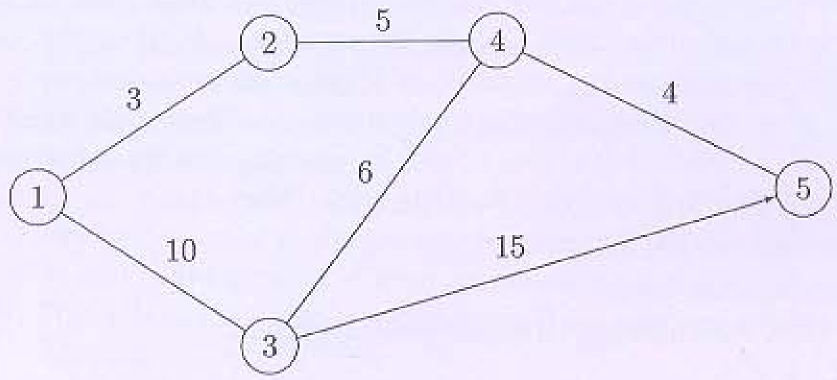
i. Queue length
ii. Traffic intensity
iii. Kandall's notation used to describe the main characteristics of queuing model.

(b) At a certain petrol pump, customers arrive according to poisson process with an average time of 5 minutes between successive arrivals. The time taken at the petrol pump to serve customers follows exponential distribution with an average of 2 minutes. Determine, [07]

i. probability that there are four customers in the system.
ii. probability that there are more than four customers in system.
iii. expected queue length.
iv. expected number of customers in the system.
v. expected time that a customer has to wait in the queue.
vi. expected time that a customer has to spend in the system.

Q.6. Answer the following questions. [14]

(a) For the network given below, find the shortest routes between every two nodes. The distances(in miles)are given on the arcs. Arc(3,5) is directional, so that no traffic is allowed from node 5 to node 3. All the other arcs allow two-way traffic. Write name of algorithm used. [07]



(b) With the help of a single server queuing model having inter-arrival and service times constantly 2 minutes and 3 minutes respectively, explain discrete simulation technique taking 15 minutes as the simulation period. Find from this average waiting time of the customer. Assume that initially the system is empty and the first customer arrives at time t = 0. [07]

OR

Q.6. Answer the following questions. [14]

(a) A small project is composed of seven activities whose time estimates are listed in the table as follows. [07]

Activity	Preceding Activity	t_o	t_m	t_p
A	-	1	1	7
B	-	1	4	7
C	-	2	2	8
D	A	1	1	1
E	B	2	5	14
F	C	2	5	8
G	D,E	3	6	15

i. Draw the project network.
ii. Find expected duration and variance of each activity. What is the expected project length?
iii. Calculate the variance of project.
iv. What is the probability that the project will be completed,

1) Atleast four weeks earlier than expected.

2) No more than four weeks later than expected.

Given,

z:	0.50	0.67	1.00	1.33	2.00
$P(Z \leq z)$:	0.6915	0.7486	0.8413	0.9082	0.9772

(b) Consider the $M/M/1$ queueing theory model(Poisson input, exponential service times and single server). Suppose that the values of *arrival rate* λ and *service rate* μ are $\lambda = 3$ per hour, $\mu = 5$ per hour. Perform Simulation of this model for 1 hour duration taking fixed - increment time = 0.1 hour.Estimate the average waiting time of customer in queuing system. Construct simulation table as [07]

t, time (min)	r_A	Arrival in interval?	Customer number	r_D	Departure in interval?	N(t)	Waiting time(min)

Where $N(t)$ denotes number of customers in system at time t.

Use following random numbers:

Random Number (Arrival)	Random Number (Departure)
0.096	0.665
0.569	0.842
0.764	0.224
0.492	0.552
0.950	0.590
0.610	0.041
0.145	0.394
0.484	0.387
0.350	0.003
0.430	0.129